

A Practical Guide to Recommender system

TOPIC -- RECOMMENDER SYSTEM

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and implemented at scale and worked through in technical reports and publications from 1994 onwards by Jussi Karlgren, then at SICS, and research groups led by Pattie Maes at MIT, Will Hill at Bellcore, and Paul Resnick, also at MIT, whose work with GroupLens was awarded the 2010 ACM Software Systems Award. Montaner provided the first overview of recommender systems from an intelligent agent perspective. Adomavicius provided a new, alternate overview of recommender systems. Herlocker provides an additional overview of evaluation techniques for recommender systems, and Beel et al. discussed the problems of offline evaluations. Beel et al. have also provided literature surveys on available research paper recommender systems and existing challenges.

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Content-based filtering Another common approach when designing recommender systems is content-based filtering. Content-based filtering methods are based on a description of the item and a profile of the user's preferences. These methods are best suited to situations where there is known data on an item (name, location, description, etc.), but not on the user. Content-based recommenders treat recommendation as a user-specific classification problem and learn a classifier for the user's likes and dislikes based on an item's features. In this system, keywords are used to describe the items, and a user profile is built to indicate the type of item this user likes. In other words, these algorithms try to recommend items similar to those that a user liked in the past or is examining in the present. It does not rely on a user sign-in mechanism to generate this often temporary profile. In particular, various candidate items are compared with items previously rated by the user, and the best-matching items are recommended. This approach has its roots in information retrieval and information filtering research. To create a user profile, the system mostly focuses on two types of information:

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In contrast to an engagement-based ranking system employed by social media and other digital platforms, a bridging-based ranking optimizes for content that is unifying instead of polarizing. Examples include Polis and Remesh which have been used around the world to help find more consensus around specific political

issues. Twitter has also used this approach for managing its community notes, which YouTube planned to pilot in 2024. Aviv Ovadya also argues for implementing bridging-based algorithms in major platforms by empowering deliberative groups that are representative of the platform's users to control the design and implementation of the algorithm.

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One approach to the design of recommender systems that has wide use is collaborative filtering. Collaborative filtering is based on the assumption that people who agreed in the past will agree in the future, and that they will like similar kinds of items as they liked in the past. The system generates recommendations using only information about rating profiles for different users or items. By locating peer users/items with a rating history similar to the current user or item, they generate recommendations using this neighborhood. This approach is a cornerstone for e-commerce sites that analyze the purchasing patterns of thousands of users to suggest what you might like. Collaborative filtering methods are classified as memory-based and model-based. A well-known example of memory-based approaches is the user-based algorithm, while that of model-based approaches is matrix factorization (recommender systems). A key advantage of the collaborative filtering approach is that it does not rely on machine analyzable content and therefore it is capable of accurately recommending complex items such as movies without requiring an "understanding" of the item itself. Many algorithms have been used in measuring user similarity or item similarity in recommender systems. For example, the k-nearest neighbor (k-NN) approach and the Pearson Correlation as first implemented by Allen. When building a model from a user's behavior, a distinction is often made between explicit and implicit forms of data collection. Examples of explicit data collection include the following:

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Performance measures Evaluation is important in assessing the effectiveness of recommendation algorithms. To measure the effectiveness of recommender systems, and compare different approaches, three types of evaluations are available: user studies, online evaluations (A/B tests), and offline evaluations. The commonly used metrics are the mean squared error and root mean squared error, the latter having been used in the Netflix Prize. The information retrieval metrics such as precision and recall or discounted cumulative gain (DCG) are useful to assess the quality of a recommendation method. Diversity, novelty, and coverage are also considered as important aspects in evaluation. However, many of the classic evaluation measures are highly criticized. Evaluating the

performance of a recommendation algorithm on a fixed test dataset will always be extremely challenging as it is impossible to accurately predict the reactions of real users to the recommendations. Hence any metric that computes the effectiveness of an algorithm in offline data will be imprecise. User studies are rather a small scale. A few dozens or hundreds of users are presented recommendations created by different recommendation approaches, and then the users judge which recommendations are best. In A/B tests, recommendations are shown to typically thousands of users of a real product, and the recommender system randomly picks at least two different recommendation approaches to generate recommendations. The effectiveness is measured with implicit measures of effectiveness such as conversion rate or click-through rate. Offline evaluations are based on historic data, e.g. a dataset that contains information about how users previously rated movies. The effectiveness of recommendation approaches is then measured based on how well a recommendation approach can predict the users' ratings in the dataset. While a rating is an explicit expression of whether a user liked a movie, such information is not available in all domains. For instance, in the domain of citation recommender systems, users typically do not rate a citation or recommended article. In such cases, offline evaluations may use implicit measures of effectiveness. For instance, it may be assumed that a recommender system is effective that is able to recommend as many articles as possible that are contained in a research article's reference list. However, this kind of offline evaluations is seen critical by many researchers. For instance, it has been shown that results of offline evaluations have low correlation with results from user studies or A/B tests. A dataset popular for offline evaluation has been shown to contain duplicate data and thus to lead to wrong conclusions in the evaluation of algorithms. Often, results of so-called offline evaluations do not correlate with actually assessed user-satisfaction. This is probably because offline training is highly biased toward the highly reachable items, and offline testing data is highly influenced by the outputs of the online recommendation module. Researchers have concluded that the results of offline evaluations should be viewed critically. Researchers have identified limits to the practical usefulness of accuracy-based evaluation metrics in recommender systems, sometimes described as an accuracy barrier or magic barrier. In this view, continued improvements in offline accuracy do not necessarily translate into better user experience, satisfaction, or real-world system impact. As a result, accuracy gains beyond a certain point may provide diminishing returns when evaluated solely through offline experiments.